

Ajay Lohith Kulla

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Career Objective

B.Tech graduate with strong programming skills in Python and Java, and hands-on experience building production-grade AI applications using RAG pipelines, LLM integration, vector databases, and Generative AI. Eager to contribute to AI product development in a collaborative, fast-paced engineering environment.

Skills

- **Languages:** Java, Python, JavaScript, SQL
- **Testing & Automation:** Selenium WebDriver, TestNG, JUnit, REST Assured, Postman, API Testing
- **AI & RAG:** RAG Pipelines, LLM Integration, Prompt Engineering, ChromaDB, FAISS, Embeddings, Semantic Search, Groq AI (LLaMA 3.3), Google Gemini, LangChain, Streamlit
- **Backend & Databases:** Spring Boot, RESTful APIs, PostgreSQL, MySQL, MongoDB, Redis, SQL Query Writing
- **Tools & Concepts:** Git, GitHub, Docker, (CI/CD basics), IntelliJ IDEA, VS Code

Education

- **B.Tech – Electrical and Electronics Engineering** **CGPA: 8.0**
Vignan's Institute of Information Technology, Visakhapatnam 2022–2026

Internship

Metaverse Ventures (Hyderabad, India) Jul 2026 – Sep 2026

- **Engineered a data synchronization pipeline** to sync PostgreSQL user data into **HydraDB** as LLM context, implementing deduplication and idempotent processing to maintain consistent, accurate agent memory.
- **Built event-driven ingestion workflows** using **Pipedream** to process webhook data from **Gmail, Slack, Jira, and GitHub**, transforming and persisting external application data into PostgreSQL.
- **Integrated personal-context and memory workflows** using **GetMembrane** and **Unabyss**, enabling agents to retrieve relevant user context for more accurate and personalized responses.

Elite Softech (Remote) Jun 2025 – Jul 2025

- **Developed and maintained RESTful backend APIs** using **Node.js** and **Express**, implementing server-side logic, API endpoints, request handling, and backend integrations.
- **Built dynamic and reusable UI components** using **React**, integrating frontend components with backend APIs.

Projects

Distributed Key-Value Store – Fault-Tolerant Distributed System **GitHub**

- **Designed and executed test scenarios** for distributed microservices built with **Spring Boot** and **API Gateway**, validating load balancing across 3 nodes and verifying high availability under failure conditions.
- **Performed API testing** using **Postman** and **REST Assured** to validate Kafka event-driven replication, Redis cache behavior, and PostgreSQL CRUD operations ensuring **99.8%** data consistency.
- **Logged, tracked, and verified defects** during regression and smoke testing of Docker-deployed services; validated idempotent Kafka event consumption and optimistic locking under concurrent write scenarios.
- **Tech Stack:** Java, Spring Boot, Kafka, Redis, PostgreSQL, Docker, Postman, REST Assured

StudyBot – AI-Powered PDF Chatbot (RAG) **Live | GitHub**

- **Built an end-to-end RAG pipeline** using **Python** and **LangChain** — PDF text extraction, recursive chunking (size 1000, overlap 100), **Google Gemini** embedding generation, and **FAISS** vector store with Top-K retrieval.
- **Engineered context-limited prompts** for **Google Gemini 2.5 Flash** grounding all responses strictly in retrieved document chunks; implemented session-state caching to avoid redundant embedding regeneration.
- **Deployed on Streamlit Cloud** with dynamic FAISS index rebuilding on new PDF uploads for seamless interactive document-based question answering.
- **Tech Stack:** Python, LangChain, FAISS, Google Gemini 2.5 Flash, PyPDF2, Streamlit, Prompt Engineering

Agentic RAG App – Company Knowledge Assistant **GitHub**

- **Built a Naive RAG pipeline** using **Python** and **ChromaDB**, indexing 5 enterprise documents into 120 semantically searchable chunks with source metadata.
- **Implemented semantic retrieval** with **SentenceTransformers** and cosine similarity, validating relevance across contextual and unrelated queries.
- **Engineered an Agentic RAG loop** with **Groq LLaMA 3.3** and tool calling, enabling autonomous retrieval decisions with a 3-iteration safety guard.
- **Tech Stack:** Python, ChromaDB, Groq AI, LangChain, SentenceTransformers, OpenAI SDK, Cosine Similarity